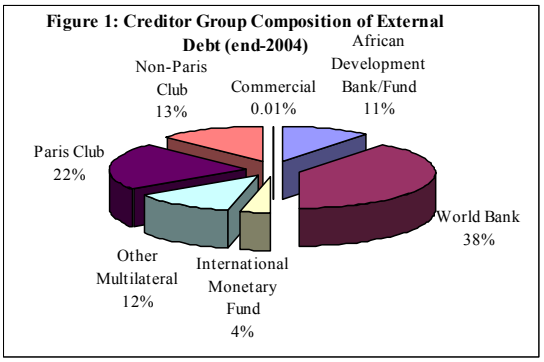
Guinea: Debt Sustainability Analysis[1]

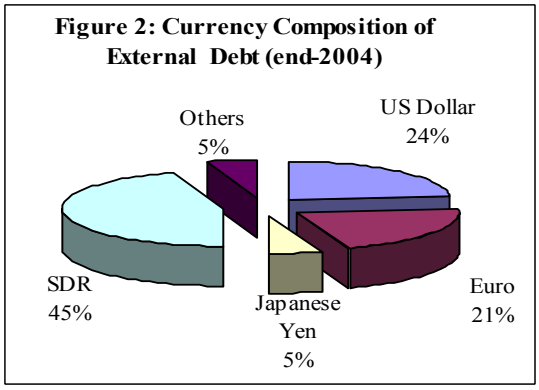
1. This annex assesses the external and public debt dynamics of Guinea using the joint IMF-World Bank debt sustainability analysis (DSA) framework for low income countries (LIC DSA). The main finding is that Guinea is in debt distress, and upon reaching completion point under the Enhanced HIPC Initiative (assumed to take place in early 2007), the country would be at a high risk of debt distress into the medium-term under the baseline scenario. Preliminary results from the public sector DSA reveal that domestic debt levels represent an additional risk to the probability of debt distress in Guinea.

Background

2. Guinea reached the decision point under the Enhanced HIPC Initiative in December 2000, qualifying for US$545 million (in NPV terms) in debt relief. Interim debt relief has been provided by all main creditors including the IMF and IDA until early 2003 when the PRGF went off-track. Only IDA and the European Union and two Paris Club creditors have continued to provide interim relief under the Initiative. Consequently, Guinea has had difficulties servicing its external debt over the period. Total external arrears to multilateral and bilateral creditors accumulated at end-2004 amounted to US$62.4 million (about 2.0 percent of GDP). HIPC debt relief will restart irrevocably once the completion point is reached.

3. **At end-2004, Guinea's public and publicly guaranteed external debt is estimated to be US$3,270 million**. Multilateral creditors account for 65 percent of the total, with the World Bank and the African Development Bank group making up three-quarters of the multilateral's share. The Paris Club group of creditors accounts for 22 percent of total, while non-Paris Club creditors and commercial creditors make up the rest. Like most HIPC countries the currency composition of its external debt leaves it vulnerable to changes in the U.S. dollar against major international currencies. With over 70 percent of Guinea's external debt denominated in euros, yen and SDR (Figure 2), the depreciation of the U.S. dollar against major international currencies since decision point has increased the U.S. dollar value of Guinea's external debt (Box 2 details the other





[1] Jointly prepared by the World Bank and IMF staff.

factors that have contributed to the increase in the NPV external debt-to-export ratio).

External Debt Sustainability Analysis (2005-25)

4. **A key feature of the low income debt sustainability analysis framework is that it compares Guinea's debt burden indicators to indicative policy-based thresholds.** The thresholds are based on the empirical finding that low-income countries with stronger policies and institutions tend to have a higher debt carrying capacity.[2] At end-2004, the NPV of debt-to-exports ratio is estimated at 191.5 percent, well above the relevant policy based indicative threshold. It is estimated that the NPV of debt-to-exports ratio would fall below the 100 percent threshold in 2015 under the baseline scenario. The NPV of debt-to-GDP and revenues ratios also breach the policy-based thresholds under the baseline scenario before declining below the thresholds in the medium-term (Table 1). The external debt service ratio quickly falls below the relevant threshold once HIPC debt relief resumes in full at completion point.

Table 1: Policy-Based External Debt Burden Indicators

	Thresholds 1/	Guinea's Ratios	
		2004	**2007-25 2/**
NPV of debt in percent of:			
Exports	100	192	97
GDP	30	40	28
Revenues	200	382	208
Debt service in percent of:			
Exports	15	21	7
Revenues 3/	20	42	5

1/ Shows policy indicative thresholds as used in the joint IMF-World Bank low-income country DSA framework for a poor policy performer.
The quality of policies and institutions is measured by the World Bank's CPIA.
2/ Simple average.
3/ Revenues excluding grants.

5. **The baseline macroeconomic framework underpinning this DSA** assumes continued implementation of sound macroeconomic policies and structural reforms under the SMP, followed by adoption of a PRGF-supported program in mid-2006, attainment of the completion point in early 2007 and no major exogenous shock or policy reversal after 2007. This scenario reflects the authorities' macroeconomic framework for 2005-08 discussed with Fund staff in the context of the 2005 Article IV consultation and conservative macroeconomic assumptions beyond 2008 (Box 1). It also assumes that all new loans will be on concessional terms.

6. **Guinea's debt burden indicators are sensitive to exogenous shocks and policy reversal.** Sensitivity analysis indicates that the baseline debt burden indicators are most vulnerable to an exogenous shock to export growth and a one-time depreciation of the Guinean franc against the U.S. dollar. The debt ratios exhibit a significant level increase and are slower to return to policy-based thresholds when exports are subjected to a one standard deviation shock for two years and a one-time franc/U.S. dollar exchange rate depreciation (see extreme stress tests in Figure 3). A scenario where key macroeconomic variables remain at historical values sharply increases the risk of debt distress. Guinea's

[2] See IDA and IMF, "Operational Framework for Debt Sustainability Assessments in Low-Income Countries—Further Considerations" (IDA/R2005-0056), April 2005.

external debt ratios would follow a continuous upward path if key economic variables remain at their historical averages from 2005.[3]

7. **An alternative scenario that simulates possible G-8 debt relief indicates a sharp reduction in the risk of debt distress at the expected completion point** (Figure 3, MDRI Scenario). The G-8 debt relief proposal calls for the write-down of debt outstanding to IDA, the IMF and the African Development Fund as of end-2004 for all HIPC countries that have reached or will reach completion point. Simulating the implementation of G-8 debt relief in 2007 reduces Guinea's external debt ratios in half on average and the ratios remain well below the policy-based thresholds for the entire projection period. **The analysis underscores the importance of the authorities' remaining committed to policy reform to achieve sustained economic growth, to reach completion point and to potentially benefit from G-8 debt relief.**

Box 1. Guinea: Macroeconomic Assumptions Underlying the DSA

Real GDP is expected to grow by 3.0 percent in 2005 and by an annual average of 5.5 percent over 2006-10 and 5.0 percent over 2011-2025. The higher growth during 2006-08 over the historically observed trend is anticipated on account of better macro policies, rebound in the mining sector and improved public infrastructure.

Average **CPI Inflation,** which was 17.5 percent in 2004, is expected to reach 30.7 percent in 2005 reflecting mainly the depreciation of the Guinean franc and the increase in petroleum prices Thereafter it is expected to decline and average 9.6 percent in 2006-08 and 5.0 percent for the remaining forecast period, based on the authorities' commitment to abandon their accommodative policy stance and to strengthen their monetary policy. Following the liberalization of the foreign exchange market in March 2005, the Guinean franc lost almost one-third of its nominal value against the US dollar. Total nominal depreciation for 2005 is expected to reach 38 percent in foreign currency terms in 2005 and to remain equal to the projected inflation differential between Guinea and its main trade partners.

Export growth is projected to average about 7.2 percent over the 2005-10 period and 8.2 percent over 2011-25.and **external current account deficit**, excluding official transfers is forecast to decline from 5.8 percent of GDP in 2004 to 4.6 percent in 2005 and 2.5, on average, over the 2006-25 period.

Fiscal policy is assumed to remain tightened in line with ongoing staff-monitored program and possible PRGF-supported program starting in mid-2006. Government revenues which averaged 11.0 percent of GDP during 1995-04 are anticipated higher at about 13 percent of GDP in 2005, 13.2 percent in 2006-08, 14.7 percent in 2009-15 and 17.2 percent in 2016-25. The increase in revenues is premised on envisaged measures to strengthen customs and tax administration, eliminate ad hoc tax or customs tariff exemptions and consolidate the implementation of the new tariff system adopted in early 2005. The primary surplus is projected to decrease from an average of 3.8 percent of GDP in 2006-08 to 2.9 percent of GDP in 2016-25.

New borrowing, the financing gap is filled mainly with concessional funds, significantly from IDA.

[3] The historical values are calculated over the 1998-04 period.

Box 2. HIPC DSA Update and Comparison to the LIC DSA Framework

This box updates the DSA presented in Guinea's decision point document (www.imf.org) using the HIPC methodology. Updates for new borrowing, exchange and discount rate parameters as of end-2004 and export data are used, while HIPC debt relief simulations conducted at decision point are redone to reflect the interruption of relief during the interim period and the accumulation of arrears.

As of end-2004, the NPV of debt after assuming the full delivery of enhanced HIPC assistance is estimated to be US$1,521 compared with US$1,293 projected for end-2004 in the decision point document. While new borrowing was lower-than-projected at decision point, it nevertheless did contribute to the increase in the NPV of external debt. Moreover, the fall in the U.S dollar vis-à-vis major international currencies and the decline in discount rates also contributed to the increase. The NPV of external debt-to-exports ratio is estimated at 189 percent as of end-2004, almost 40 percentage points higher than projected at decision point. Export growth between decision point and end-2004 fell well short of expectations. However, the export growth that was realized between decision point and end-2004 contributed to a small reduction in the ratio, but the reduction was far from sufficient to offset the increase in the ratio due to new borrowing and the change in exchange and discount rates

Guinea: Change in the NPV of Debt-to-Exports Ratio from Decision Point to end-2004 1/

	Anticipated Change in Ratio from end-1999 to end-2004 2/	Unanticipated Change in the end-2004 Ratio	Total Change in Ratio
	Percentage points		
End-1999 NPV of debt-to-exports ratio	**150.0**		**150.0**
End-2004 NPV of debt-to-exports ratio		**118.6 2/**	
Factors contributing to changes in ratios			
New borrowing (2000-04)	14.0	-4.3	9.7
Export growth (2000-04)	-53.6	49.4	-4.1
Changes in parameters		25.3	25.3
of which			
Due to changes in the discount rates		11.7	11.7
Due to changes in the exchange rates		13.6	13.6
Other factors 3/		8.4	8.4
End-2004 NPV of debt-to-exports ratio	**118.6**	**189.0 4/**	**189.0 4/**

Source: Staff estimates.

1/ All figures assume the full delivery of debt relief under the HIPC Initiative.

2/ End-1999 data and projections at the time of the decision point.

3/ Includes accumulation of arrears, changes to the simulated HIPC debt relief relative to the decision point and a historical revision to exports.

4/ Based on actual end-2003 debt stocks, discount rates, exchange rates and the three-year backward-looking average of exports of good and services.

Using the HIPC methodology to update the decision point DSA results in an end-2004 ratio (189 percent) that is slightly lower than that generated with the new LIC DSA framework (192 percent). There are a number of methodological differences between the two frameworks that explain the difference: (i) the LIC DSA uses annual exports rather than a three-year backward looking average; (ii) the LIC DSA uses exchange rate projections from the WEO to project debt service streams rather than end-2004 exchanges rates under the HIPC methodology; and (iii) the LIC DSA uses a single discount rate of 5 percent to calculate the NPV of debt rather than currency specific discount rates.

In the case of Guinea, the difference resulting from the change in the exchange rates used to project future debt streams increases the debt ratios by more than the offsetting changes in discount rates and exports used between the two methods.

Public Sector Debt Sustainability Analysis (2005-25)

8. **Guinea's central government fiscal deficit (commitment basis, including grants) has been at an average deficit of 6.3 per cent of GDP during the period 1994 to 2004**. Reflecting fiscal slippages, the deficit deteriorated steadily from 5.3 percent of GDP in 1999 to reach 8.9 percent of GDP by end-2003. The fiscal imbalances reflect higher-than-budgeted expenditures on account of defense spending, interest charges on domestic debt, and domestically financed investment outlays. During the period, expenditures averaged 17.2 percent of GDP while revenues (including grants) were 13.7 percent of GDP on average, resulting in the growing primary deficit and the accumulation of domestic debt.

9. **At end-2004, the government's domestic debt represented about 22 percent of GDP** and included advances from the central bank (15 percent), treasury bills owned by commercial banks (4 percent) and accumulated arrears to the private sector (3 percent). The government had also a sizable amount of non-certified debt to the private non-financial sector and public enterprises, which is awaiting certification by external auditors. This category of debt is not captured in the data used in this analysis.[4] In the context of ongoing Staff-Monitored Program, the authorities have undertaken an inventory of its debt vis-à-vis the private non-financial sector and public enterprises, and intend to complete an audit of this debt by end-December 2005.

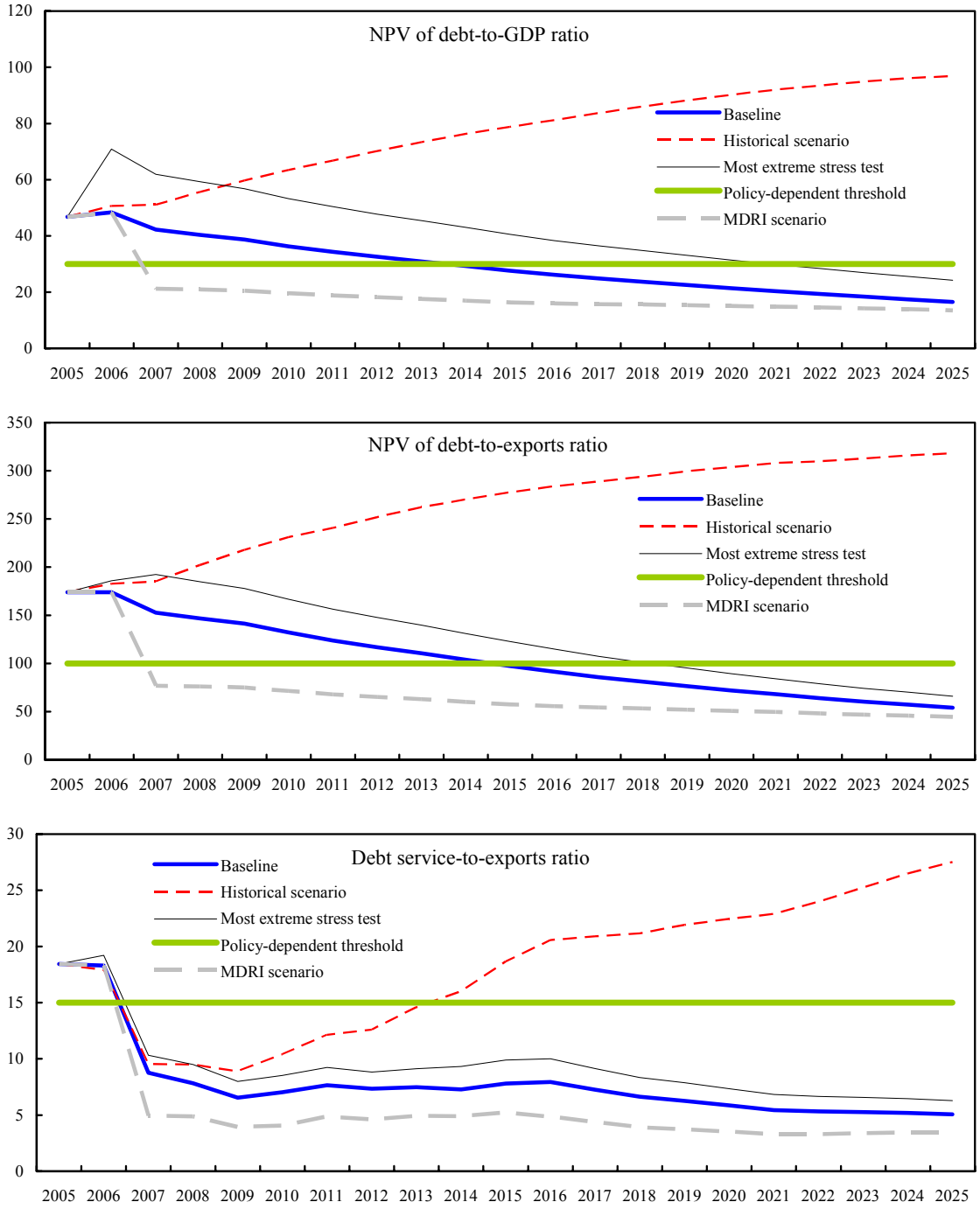
10. **An important component of the baseline macroeconomic scenario is increasing government revenues combined with restrained government spending.** The resulting fiscal adjustment is expected to reduce the public sector borrowing requirement over time. Consequently, the domestic debt stock is assumed to fall over time, reaching 7 percent of GDP by the end of the projection period. The combination of strong revenue growth and a diminishing domestic debt stock, result in a falling public sector debt burden over time. However, the starting levels are extremely high with the NPV of total public sector debt to fiscal revenues ratio estimated to be 577 percent and the NPV of public debt to GDP to be 134 percent in 2005 (Table 4).

11. **As in the external debt sustainability analysis, sensitivity analysis indicates a sharp rise in the risk of debt over the medium term**. In the no reform scenario, which maintains the primary balance at the 2004 level, all debt burden indicators increase significantly relative to the baseline scenario. For example, the NPV of public debt to revenues ratio increases on average about 130 percentage points. A one-time 30 percent real depreciation in the franc would also generate a significant increase in the debt burden indicators (most extreme stress tests in Figure 4).

[4] One of the stress tests in the LIC DSA framework assumes a 10 percent of GDP increase in debt creating flows in 2006, which could approximate the inclusion of non-certified domestic debt.

12. **The results of the public sector DSA underscore the importance of domestic debt management, especially the limiting of domestic borrowing from the central bank and the accumulation of domestic arrears to the private sector.**

Figure 3. Guinea: Indicators of Public and Publicly Guaranteed External Debt
Under Alternative Scenarios, 2005-2025
(In percent)



Source: Staff projections and simulations.

Table 2. Guinea: External Debt Sustainability Framework, Baseline Scenario, 2005-2025 1/
(In percent of GDP, unless otherwise indicated)

	Actual	Historical Average 6/	Standard Deviation 6/	Projections								
	2004			2005	2006	2007	2008	2009	2010	2005-10 Average	2015	2025
External debt (nominal) 1/	**82.4**			**97.4**	**99.5**	**93.3**	**87.4**	**81.7**	**75.3**	**89.1**	**54.1**	**30.3**
o/w public and publicly guaranteed (PPG)	82.4			97.4	99.5	93.3	87.4	81.7	75.3		54.1	30.3
Change in external debt	-10.2			15.1	2.1	-6.2	-5.9	-5.6	-6.4		-3.7	-1.8
Identified net debt-creating flows	-6.0			-3.4	-5.2	-5.6	-5.6	-4.1	-4.1		-1.7	0.1
Non-interest current account deficit	**4.2**	**4.3**	**1.7**	**2.3**	**2.7**	**2.5**	**1.7**	**2.0**	**2.0**	**2.2**	**2.4**	**2.8**
Deficit in balance of goods and services	4.1			2.6	3.1	2.1	1.4	1.8	1.9		2.3	2.6
Exports	20.8			26.9	27.8	27.6	27.5	27.4	27.4	**27.4**	28.4	30.5
Imports	24.8			29.5	30.8	29.8	28.9	29.2	29.4		30.7	33.1
Net current transfers (negative = inflow)	-0.1	-0.4	0.7	-0.3	-0.3	0.1	0.0	-0.1	-0.3		-0.1	0.0
Other current account flows (negative = net inflow)	0.2			-0.1	-0.1	0.2	0.3	0.4	0.4		0.3	0.1
Net FDI (negative = inflow)	**-3.6**	**-0.7**	**1.8**	**-4.2**	**-4.4**	**-3.9**	**-2.8**	**-2.3**	**-2.6**		**-2.0**	**-1.6**
Endogenous debt dynamics 2/	**-6.6**			**-1.5**	**-3.5**	**-4.2**	**-4.4**	**-3.8**	**-3.6**		**-2.1**	**-1.1**
Contribution from nominal interest rate	1.3			1.5	1.4	0.8	0.8	0.7	0.6		0.5	0.4
Contribution from real GDP growth	-2.3			-3.0	-4.9	-5.0	-5.3	-4.5	-4.2		-2.7	-1.5
Contribution from price and exchange rate changes	-5.7			…	…	…	…	…	…		…	...
Residual (3-4) 3/	**-4.2**			**18.5**	**7.3**	**-0.6**	**-0.3**	**-1.5**	**-2.3**		**-2.0**	**-1.9**
o/w exceptional financing	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0
NPV of external debt 4/	39.7			46.8	48.3	42.2	40.4	38.7	36.3	**42.1**	27.6	16.5
In percent of exports	191.5			173.9	174.1	152.7	146.8	141.3	132.2	**153.5**	97.4	54.1
NPV of PPG external debt	**39.7**			**46.8**	**48.3**	**42.2**	**40.4**	**38.7**	**36.3**	**42.1**	**27.6**	**16.5**
In percent of exports	191.5			173.9	174.1	152.7	146.8	141.3	132.2	**153.5**	97.4	54.1
Debt service-to-exports ratio (in percent)	21.0			18.4	18.3	8.8	7.8	6.6	7.0	**11.2**	7.8	5.1
PPG debt service-to-exports ratio (in percent)	21.0			18.4	18.3	8.8	7.8	6.6	7.0	**11.2**	7.8	5.1
Total gross financing need (billions of U.S. dollars)	0.2			0.1	0.1	0.0	0.0	0.1	0.1	**0.1**	0.2	0.4
Non-interest current account deficit that stabilizes debt ratio	14.4			-12.8	0.6	8.7	7.6	7.7	8.5	**3.4**	6.1	4.6
Key macroeconomic assumptions												
Real GDP growth (in percent)	2.7	3.3	1.4	3.0	5.0	5.4	6.1	5.5	5.5	5.1	5.0	5.0
GDP deflator in US dollar terms (change in percent)	6.5	-2.2	8.9	-19.2	-5.9	2.8	2.4	2.7	3.4	-2.3	2.3	1.9
Effective interest rate (percent) 5/	1.6	1.4	0.3	1.5	1.4	0.9	1.0	0.8	0.8	1.1	1.0	1.4
Growth of exports of G&S (US dollar terms, in percent)	2.8	1.6	4.5	7.9	1.9	7.9	8.1	7.9	9.3	7.2	7.9	7.1
Growth of imports of G&S (US dollar terms, in percent)	7.1	2.1	6.8	-1.0	3.1	4.7	5.4	9.5	9.8	5.3	8.4	7.5
Grant element of new public sector borrowing (in percent)	...	...	...	29.2	22.8	33.1	38.6	39.0	45.0	34.6	38.8	36.3
Memorandum item:												
Nominal GDP (billions of US dollars)	4.0			3.3	3.3	3.5	3.8	4.2	4.5		6.5	13.4

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

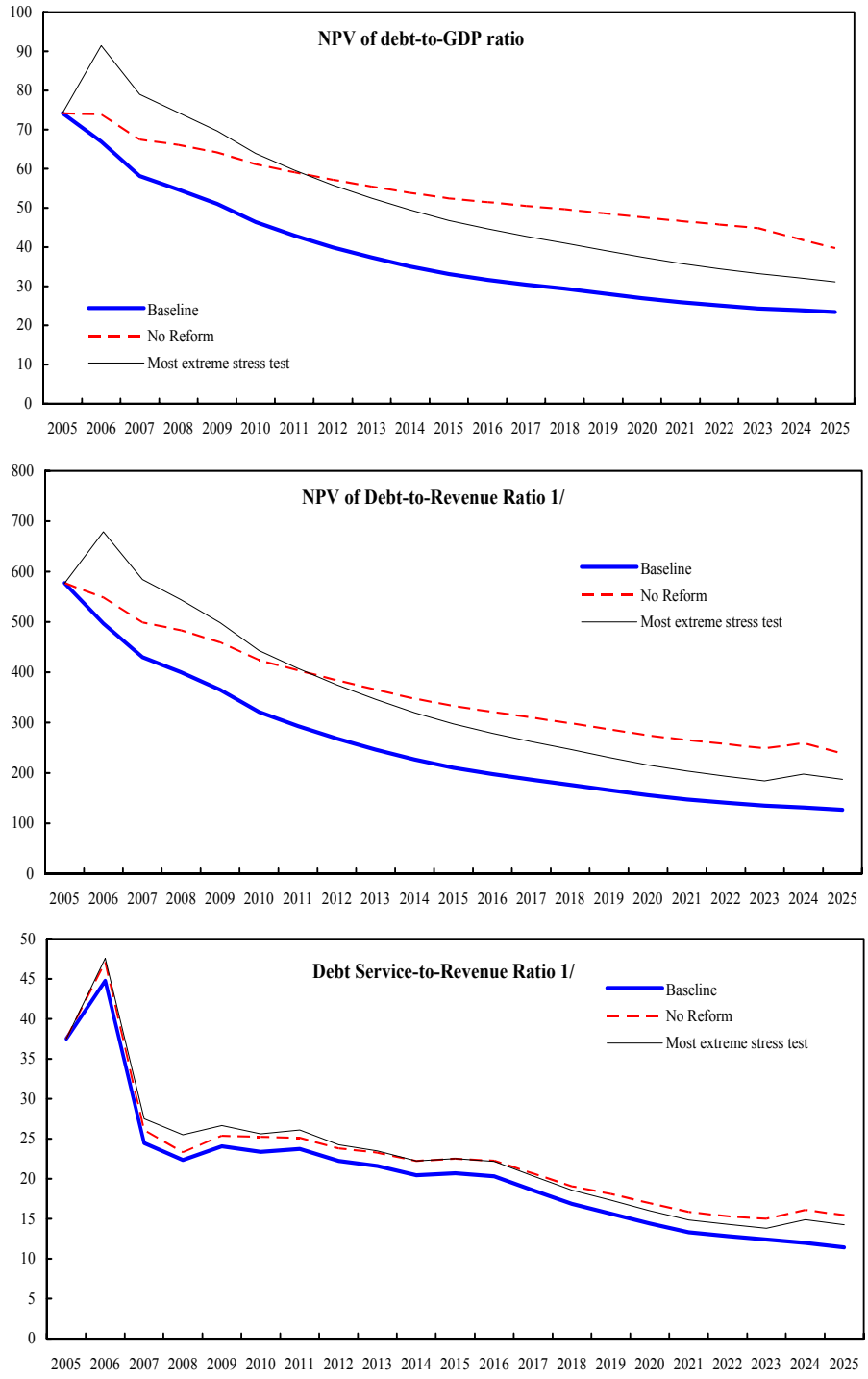
Table 3. Guinea: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2005-25
(In percent)

		Projections						
	2005	2006	2007	2008	2009	2010	**2015**	2025
NPV of debt-to-GDP ratio								
Baseline	47	48	42	40	39	36	**28**	16
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2005-25 1/	47	51	51	56	60	63	**79**	97
A2. New public sector loans on less favorable terms in 2005-25 2/	47	49	44	42	41	39	**32**	24
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	47	50	45	43	41	39	**29**	18
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	47	49	46	44	42	39	**30**	17
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	47	51	52	49	47	44	**34**	20
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	47	52	50	47	45	43	**32**	18
B5. Combination of B1-B4 using one-half standard deviation shocks	47	50	52	50	48	45	**34**	20
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	47	71	62	59	57	53	**41**	24
NPV of debt-to-exports ratio								
Baseline	174	174	153	147	141	132	**97**	54
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2005-25 1/	174	183	185	202	218	231	**278**	318
A2. New public sector loans on less favorable terms in 2005-25 2/	174	177	158	153	149	142	**114**	78
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	174	174	153	147	141	132	**97**	54
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	174	186	192	185	178	167	**123**	66
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	174	174	153	147	141	132	**97**	54
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	174	189	179	172	166	156	**114**	59
B5. Combination of B1-B4 using one-half standard deviation shocks	174	181	180	173	166	156	**115**	62
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	174	174	153	147	141	132	**97**	54
Debt service-to-exports ratio								
Baseline	18	18	9	8	7	7	**8**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2005-25 1/	18	18	10	9	9	10	**19**	28
A2. New public sector loans on less favorable terms in 2005-25 2/	18	18	9	8	7	7	**8**	6
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2006-07	18	18	9	8	7	7	**8**	5
B2. Export value growth at historical average minus one standard deviation in 2006-07 3/	18	19	10	9	8	9	**10**	6
B3. US dollar GDP deflator at historical average minus one standard deviation in 2006-07	18	18	9	8	7	7	**8**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2006-07 4/	18	18	9	8	7	8	**9**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	18	19	10	9	8	8	**9**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2006 5/	18	18	9	8	7	7	**8**	5
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	35	35	35	35	35	35	**35**	35

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Figure 4. Guinea: Indicators of Public Debt Under Alternative Scenarios, 2005-25



Source: Staff projections and simulations.
1/ Revenue including grants.

Table 4. Guinea: Public Sector Debt Sustainability Framework, Baseline Scenario, 2005-25
(in percent of GDP, unless otherwise indicated)

	Actual			Estimate		Projections						
	2004	Historical Average 5/	Standard Deviation 5/	2005	2006	2007	2008	2009	2010	2005-10 Average	2015	2025
Public sector debt 1/	114.4			133.8	119.5	109.7	102.2	94.5	85.9		59.9	37.4
o/w foreign-currency denominated	95.2			117.0	104.3	96.1	89.9	84.0	77.3		55.3	31.4
Change in public sector debt	-0.7			19.4	-14.3	-9.8	-7.5	-7.7	-8.6		-4.0	-1.3
Identified debt-creating flows	2.3			21.6	-16.2	-13.0	-10.3	-9.3	-9.2		-4.2	0.9
Primary deficit	2.4	2.0	1.4	-1.4	-1.9	-1.2	-1.1	-1.0	-1.1	-1.3	-0.3	2.8
Revenue and grants	11.4			12.9	13.5	13.5	13.7	14.0	14.4		15.8	16.6
of which : grants	1.0			0.8	0.7	0.2	0.2	0.2	0.2		0.2	0.1
Primary (noninterest) expenditure	13.8			11.5	11.6	12.4	12.5	13.0	13.4		15.5	19.4
Automatic debt dynamics	0.5			23.4	-13.9	-9.7	-7.3	-6.6	-6.7		-3.3	-1.8
Contribution from interest rate/growth differential	-6.3			-6.8	-7.9	-7.2	-7.0	-6.1	-5.7		-3.2	-1.8
of which : contribution from average real interest rate	-3.3			-3.5	-1.6	-1.1	-0.6	-0.7	-0.7		-0.1	0.0
of which : contribution from real GDP growth	-3.0			-3.3	-6.3	-6.1	-6.3	-5.4	-5.0		-3.0	-1.9
Contribution from real exchange rate depreciation	6.8			30.2	-6.0	-2.4	-0.3	-0.5	-1.1		...	...
Other identified debt-creating flows	-0.6			-0.5	-0.4	-2.2	-1.9	-1.7	-1.4		-0.6	0.0
Privatization receipts (negative)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0
Recognition of implicit or contingent liabilities	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0
Debt relief (HIPC and other)	-0.5			-0.5	-0.4	-2.2	-1.9	-1.7	-1.4		-0.6	0.0
Other (specify, e.g. bank recapitalization)	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0
Residual, including asset changes	-3.0			-2.2	1.8	3.2	2.8	1.6	0.5		0.2	-2.3
NPV of public sector debt	66.5			74.2	66.9	58.1	54.7	51.0	46.4		33.1	23.4
o/w foreign-currency denominated	47.3			57.4	51.7	44.5	42.5	40.5	37.8		28.5	17.5
o/w external	44.7			55.1	49.6	42.6	40.8	39.1	36.6		27.9	16.6
NPV of contingent liabilities (not included in public sector debt)	...			...	...	...	...	...	...		...	...
Gross financing need 2/	8.9			4.3	4.9	2.4	2.2	2.6	2.5		3.2	3.1
NPV of public sector debt-to-revenue ratio (in percent) 3/	582.8			577.1	496.8	429.8	400.0	365.1	321.3		210.1	140.8
o/w external	391.5			428.2	368.5	315.1	298.2	279.7	253.6		177.1	100.1
Debt service-to-revenue ratio (in percent) 3/ 4/	48.3			37.5	44.8	24.5	22.4	24.1	23.4		20.7	11.4
Primary deficit that stabilizes the debt-to-GDP ratio	3.0			-20.7	12.5	8.6	6.4	6.7	7.6		3.7	4.1
Key macroeconomic and fiscal assumptions												
Real GDP growth (in percent)	2.7	3.7	1.4	3.0	5.0	5.4	6.1	5.5	5.5	5.1	5.0	5.0
Average nominal interest rate on forex debt (in percent)	2.0	0.3	0.7	2.2	1.5	1.0	1.1	1.0	0.9	1.3	1.1	1.6
Average real interest rate on domestic currency debt (in percent)	-13.0	-8.1	3.9	-17.1	-6.3	-0.9	1.5	1.5	1.5	-3.3	7.4	2.8
Real exchange rate depreciation (in percent, + indicates depreciation)	7.6	6.6	11.1	32.8	-5.4	-2.5	-0.3	-0.6	-1.3	4.6	-0.3	0.1
Inflation rate (GDP deflator, in percent)	21.7	6.7	6.5	28.4	12.9	6.7	4.4	4.7	5.5	10.4	4.4	3.9
Growth of real primary spending (deflated by GDP deflator, in percent)	-18.0	1.7	13.0	-13.9	5.9	12.3	7.6	9.5	8.5	5.0	7.4	7.0

Sources: Country authorities; and Fund staff estimates and projections.

1/ Public sector debt refers to central government and nonfinancial public sector on a gross basis.

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues including grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

6/ The implicit nominal interest rate on domestic debt differs from nominal effective rates because the implicit rates are calculated on the basis of the total stock of debt to the central bank, commercial banks, and the private non-financial sector, which is higher than the stock of domestic debt which is actually being serviced.

Table 5. Guinea: Sensitivy Analysis for Key Indicators of Public Debt, 2005-25

	Estimate	Projections						
	2005	2006	2007	2008	2009	2010	2015	2025
NPV of Debt-to-GDP Ratio								
Baseline	74	67	58	55	51	46	33	23
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	74	71	66	66	65	63	56	44
A2. Primary balance is unchanged from 2004	78	74	67	66	64	61	52	40
A3. Permanently lower GDP growth 1/	74	67	59	55	52	48	36	30
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	74	69	63	60	56	52	40	32
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	74	72	67	62	58	52	37	25
B3. Combination of B1-B2 using one half standard deviation shocks	74	72	68	64	59	54	38	26
B4. One-time 30 percent real depreciation in 2006	74	91	79	74	70	64	47	31
B5. 10 percent of GDP increase in other debt-creating flows in 2006	74	76	66	62	57	52	37	25
NPV of Debt-to-Revenue Ratio 2/								
Baseline	577	497	430	400	365	321	210	127
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	577	529	488	483	465	435	352	261
A2. Primary balance is unchanged from 2004	603	548	499	484	459	424	333	239
A3. Permanently lower GDP growth 1/	577	499	434	406	373	330	229	183
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	577	512	462	435	402	359	253	194
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	577	533	493	454	412	362	234	151
B3. Combination of B1-B2 using one half standard deviation shocks	577	538	503	465	422	371	241	154
B4. One-time 30 percent real depreciation in 2006	577	679	584	544	498	443	297	187
B5. 10 percent of GDP increase in other debt-creating flows in 2006	577	566	490	452	410	360	234	151
Debt Service-to-Revenue Ratio 2/								
Baseline	38	45	24	22	24	23	21	11
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	47	46	26	24	27	27	25	19
A2. Primary balance is unchanged from 2004	47	47	26	23	25	25	23	15
A3. Permanently lower GDP growth 1/	47	45	25	23	24	24	21	14
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2006-2007	47	46	26	24	26	25	22	14
B2. Primary balance is at historical average minus one standard deviations in 2006-2007	47	45	26	23	25	24	21	13
B3. Combination of B1-B2 using one half standard deviation shocks	47	46	26	24	26	25	22	14
B4. One-time 30 percent real depreciation in 2006	47	48	28	25	27	26	23	14
B5. 10 percent of GDP increase in other debt-creating flows in 2006	47	45	27	23	25	24	21	13

Sources: Country authorities; and Fund staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of 20 (i.e., the length of the projection period).
2/ Revenues are defined inclusive of grants.